Roll No. | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-6)

MEDICINAL CHEMISTRY-III-THEORY

Subject Code : BP-601T

M.Code : 77986

Date of Examination:13-12-2025

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carryingTEN marks each and students have to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.**

SECTION-A

1. Answer the following :

a) Write the structure and uses of Ampicillin.

b) Give synthesis of mebendazole.

c) Give the structure and uses of any one antiviral agent.

d) Define Hansch analysis and mention its significance in drug design.

e) State any two advantages of designing prodrugs.

f) Mention the mechanism of action of Isoniazid.

g) Write the structure and uses of Miconazole.

h) Draw the basic structure of tetracycline and mention any two derivatives.

i) Mention the steric parameters used in QSAR.

j) Write the SAR of 4-aminoquinolines with respect to antimalarial activity.

SECTION-B

2. What are beta lactam antibiotics? Give the degradation products of penicillins and cephalosporins. Write a note on beta lactamase inhibitors.

3. What are antimalarial agents? Classify them with example. Give the MOA and synthesis of quinolones and Primaquine.

4. What are antifungal agents? Classify them with examples and write in detail about polyene antifungal agents.

SECTION-C

5. Write the synthesis and MOA of Chloramphenicol.

6. What areanti-TB drugs? Enlist the problems associated with the treatment. Give the synthesis of INH.

7. Discuss miscellaneous UTIs. Outline the synthesis of ciprofloxacin.

8. Define prodrugs. Describe any two examples highlighting the rationale behind their design.

9. Explain the concept of QSAR. Discuss the role of physicochemical parameters such as partition coefficient and electronic effects in drug design.

10. Describe the chemistry, classification and SAR of sulfonamides. Mention at least three examples with their therapeutic uses.

11. Classify antiviral agents based on their mechanism of action and write a short note on Acyclovir.

12. Discuss the Structure Activity Relationship (SAR) of Penicillin and explain how β-lactam ring modification affects its activity.

13. Write the structure and uses of Clindamycin and Ciprofloxacin.

of Pharmacy

Nov. 2025

NOTE : Disclosure of identity by writing mobile number or making passing request on any page of Answer sheet will lead to UMC against the Student.